

10

One-Sample Tests of Hypothesis



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- ▲ **DOLE PINEAPPLE INC.** is concerned that the 16-ounce can of sliced pineapple is being overfilled. Assume the standard deviation of the process is .03 ounce. The quality control department took a random sample of 50 cans and found that the arithmetic mean weight was 16.05 ounces. At the 5% level of significance, can we conclude that the mean weight is greater than 16 ounces? Determine the p -value. (See Exercise 26 and **LO10-4**.)

LEARNING OBJECTIVES

When you have completed this chapter, you will be able to:

- LO10-1** Explain the process of testing a hypothesis.
- LO10-2** Apply the six-step procedure for testing a hypothesis.
- LO10-3** Distinguish between a one-tailed and a two-tailed test of hypothesis.
- LO10-4** Conduct a test of a hypothesis about a population mean.
- LO10-5** Compute and interpret a p -value.
- LO10-6** Use a t statistic to test a hypothesis.
- LO10-7** Compute the probability of a Type II error.

INTRODUCTION

Chapter 8 began our study sampling and statistical inference. We described how we could select a random sample to estimate the value of a population parameter. For example, we selected a sample of five employees at Spence Sprockets, found the number of years of service for each sampled employee, computed the mean years of service, and used the sample mean to estimate the mean years of service for all employees. In other words, we estimated a population parameter from a sample statistic.

Chapter 9 continued the study of statistical inference by developing a confidence interval. A confidence interval is a range of values within which we expect the population parameter to occur. In this chapter, rather than develop a range of values within which we expect the population parameter to occur, we develop a procedure to test the validity of a statement about a population parameter. Some examples of statements we might want to test are:

- The mean speed of automobiles passing milepost 150 on the West Virginia Turnpike is 68 miles per hour.
- The mean number of miles driven by those leasing a Chevy Trail-Blazer for 3 years is 32,000 miles.
- The mean time an American family lives in a particular single-family dwelling is 11.8 years.
- In 2016, the mean starting salary for a graduate from a four-year business program is \$51,541.
- According to the Kelley Blue Book (www.kbb.com), a 2017 Ford Edge averages 21 miles per gallon in the city.
- The mean cost to remodel a kitchen is \$20,000.



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This chapter and several of the following chapters cover statistical hypothesis testing. We begin by defining what we mean by a statistical hypothesis and statistical hypothesis testing. Next, we outline the steps in statistical hypothesis testing. Then we conduct tests of hypothesis for means. In the last section of the chapter, we describe possible errors due to sampling in hypothesis testing.

LO10-1

Explain the process of testing a hypothesis.

WHAT IS HYPOTHESIS TESTING?

The terms *hypothesis testing* and *testing a hypothesis* are used interchangeably. Hypothesis testing starts with a statement, or assumption, about a population parameter—such as the population mean. This statement is referred to as a hypothesis.

HYPOTHESIS A statement about a population parameter subject to verification.

A hypothesis might be that the mean monthly commission of sales associates in retail electronics stores, such as hhgregg, is \$2,000. We cannot contact all hhgregg sales associates to determine that the mean is \$2,000. The cost of locating and interviewing every hhgregg electronics sales associate in the United States would be exorbitant. To test the validity of the hypothesis ($\mu = \$2,000$), we must select a sample from the population of all hhgregg electronics sales associates, calculate sample statistics,

and based on certain decision rules reject or fail to reject the hypothesis. A sample mean of \$1,000 per month is much less than \$2,000 per month and we would most likely reject the hypothesis. However, suppose the sample mean is \$1,995. Can we attribute the \$5 difference between \$1,995 and \$2,000 to sampling error? Or is this difference of \$5 statistically significant?

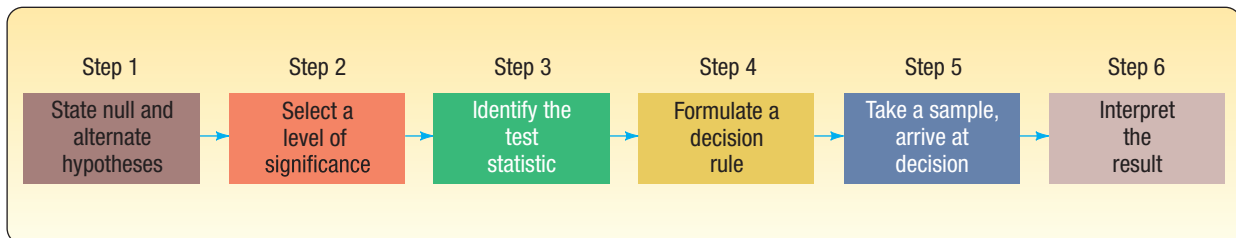
HYPOTHESIS TESTING A procedure based on sample evidence and probability theory to determine whether the hypothesis is a reasonable statement.

LO10-2

Apply the six-step procedure for testing a hypothesis.

SIX-STEP PROCEDURE FOR TESTING A HYPOTHESIS

There is a six-step procedure that systematizes hypothesis testing; when we get to step 6 we are ready to interpret the results of the test based on the decision to reject or not reject the hypothesis. However, hypothesis testing as used by statisticians does not provide proof that something is true, in the manner in which a mathematician “proves” a statement. It does provide a kind of “proof beyond a reasonable doubt,” in the manner of the court system. Hence, there are specific rules of evidence, or procedures, that are followed. The steps are shown in the following diagram. We will discuss in detail each of the steps.



Step 1: State the Null Hypothesis (H_0) and the Alternate Hypothesis (H_1)

The first step is to state the hypothesis being tested. It is called the **null hypothesis**, designated H_0 , and read “*H sub zero*.” The capital letter H stands for hypothesis, and the subscript zero implies “no difference.” There is usually a “not” or a “no” term in the null hypothesis, meaning that there is “no change.” For example, the null hypothesis is that the mean number of miles driven on the steel-belted tire is not different from 60,000. The null hypothesis would be written $H_0: \mu = 60,000$. Generally speaking, the null hypothesis is developed for the purpose of testing. We either reject or fail to reject the null hypothesis. The null hypothesis is a statement that is not rejected unless our sample data provide convincing evidence that it is false.

We should emphasize that, if the null hypothesis is not rejected on the basis of the sample data, we cannot say that the null hypothesis is true. To put it another way, failing to reject the null hypothesis does not prove that H_0 is true; it means we have *failed to disprove* H_0 . To prove without any doubt the null hypothesis is true, the population parameter would have to be known. To actually determine it, we would have to test, survey, or count every item in the population. This is usually not feasible. The alternative is to take a sample from the population.

Often, the **null hypothesis** begins by stating, “There is no *significant* difference between . . .” or “The mean impact strength of the glass is not *significantly* different

from. . .” When we select a sample from a population, the sample statistic is usually numerically different from the hypothesized population parameter. As an illustration, suppose the hypothesized impact strength of a glass plate is 70 psi, and the mean impact strength of a sample of 12 glass plates is 69.5 psi. We must make a decision about the difference of 0.5 psi. Is it a true difference, that is, a significant difference, or is the difference between the sample statistic (69.5) and the hypothesized population parameter (70.0) due to chance (sampling)? To answer this question, we conduct a test of significance, commonly referred to as a test of hypothesis. To define what is meant by a null hypothesis:

NULL HYPOTHESIS A statement about the value of a population parameter developed for the purpose of testing numerical evidence.

The **alternate hypothesis** describes what you will conclude if you reject the null hypothesis. It is written H_1 and is read “*H sub one*.” It is also referred to as the research hypothesis. The alternate hypothesis is accepted if the sample data provide us with enough statistical evidence that the null hypothesis is false.

ALTERNATE HYPOTHESIS A statement that is accepted if the sample data provide sufficient evidence that the null hypothesis is false.

The following example will help clarify what is meant by the null hypothesis and the alternate hypothesis. A recent article indicated the mean age of U.S. commercial aircraft is 15 years. To conduct a statistical test regarding this statement, the first step is to determine the null and the alternate hypotheses. The null hypothesis represents the current or reported condition. It is written $H_0: \mu = 15$. The alternate hypothesis is that the statement is not true, that is, $H_1: \mu \neq 15$. It is important to remember that no matter how the problem is stated, *the null hypothesis will always contain the equal sign*. The equal sign (=) will never appear in the alternate hypothesis. Why? Because the null hypothesis is the statement being tested, and we need a specific value to include in our calculations. We turn to the alternate hypothesis only if the data suggest the null hypothesis is untrue.

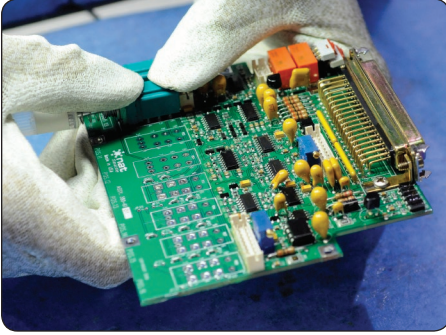
Step 2: Select a Level of Significance

After setting up the null hypothesis and alternate hypothesis, the next step is to state the level of significance.

LEVEL OF SIGNIFICANCE The probability of rejecting the null hypothesis when it is true.

The level of significance is designated α , the Greek letter alpha. It is also sometimes called the level of risk. This may be a more appropriate term because it is the risk you take of rejecting the null hypothesis when it is really true.

There is no one level of significance that is applied to all tests. A decision is made to use the .05 level (often stated as the 5% level), the .01 level, the .10 level, or any other level between 0 and 1. Traditionally, the .05 level is selected for consumer research projects, .01 for quality assurance, and .10 for political polling. You, the researcher, must decide on the level of significance *before* formulating a decision rule and collecting sample data.



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To illustrate how it is possible to reject a true hypothesis, suppose a firm manufacturing personal computers uses a large number of printed circuit boards. Suppliers bid on the boards, and the one with the lowest bid is awarded a sizable contract. Suppose the contract specifies that the computer manufacturer's quality-assurance department will randomly sample all incoming shipments of circuit boards. If more than 6% of the boards sampled are substandard, the shipment will be rejected. The null hypothesis is that the incoming shipment of boards meets the quality standards of the contract and contains 6% or less defective boards. The alternate hypothesis is that more than 6% of the boards are defective.

A shipment of 4,000 circuit boards was received from Allied Electronics, and the quality assurance department selected a random sample of 50 circuit boards for testing. Of the 50 circuit boards sampled, 4 boards, or 8%, were substandard. The shipment was rejected because it exceeded the maximum of 6% substandard printed circuit boards. If the shipment was actually substandard, then the decision to return the boards to the supplier was correct.

However, because of sampling error, there is a small probability of an incorrect decision. Suppose there were only 40, or 4%, defective boards in the shipment (well under the 6% threshold) and 4 of these 40 were randomly selected in the sample of 50. The sample evidence indicates that the percentage of defective boards is 8% (4 out of 50 is 8%) so we reject the shipment. But, in fact, of the 4,000 boards, there are only 40 defective units. The true defect rate is 1.00%. In this instance our sample evidence estimates 8% defective but there is only 1% defective in the population. Based on the sample evidence, an incorrect decision was made. In terms of hypothesis testing, we rejected the null hypothesis when we should have failed to reject the null hypothesis. By rejecting a true null hypothesis, we committed a Type I error. The probability of committing a Type I error is represented by the Greek letter alpha (α).

TYPE I ERROR Rejecting the null hypothesis, H_0 , when it is true.

The other possible error in hypothesis testing is called Type II error. The probability of committing a Type II error is designated by the Greek letter beta (β).

TYPE II ERROR Not rejecting the null hypothesis when it is false.

The firm manufacturing personal computers would commit a Type II error if, unknown to the manufacturer, an incoming shipment of printed circuit boards from Allied Electronics contained 15% substandard boards, yet the shipment was accepted. How could this happen? A random sample of 50 boards could have 2 (4%) substandard boards, and 48 good boards. According to the stated procedure, because the sample contained less than 6% substandard boards, the decision is to accept the shipment. This is a Type II error. While this event is extremely unlikely, it is possible based on the process of randomly sampling from a population. In a later section, we show how to calculate the probability of a Type II error.

In retrospect, the researcher cannot study every item or individual in the population. Thus, there is a possibility of two types of error—a Type I error, wherein the null hypothesis is rejected when it should not be rejected, and a Type II error, wherein the null hypothesis is not rejected when it should have been rejected.

We often refer to the probability of these two possible errors as *alpha*, α , and *beta*, β . Alpha (α) is the probability of making a Type I error, and beta (β) is the probability of making a Type II error. The following table summarizes the decisions the researcher could make and the possible consequences.

Null Hypothesis	Researcher	
	Does Not Reject H_0	Rejects H_0
H_0 is true	Correct decision	Type I error
H_0 is false	Type II error	Correct decision

Step 3: Select the Test Statistic

There are many **test statistics**. In this chapter, we use both z and t as the test statistics. In later chapters, we will use such test statistics as F and χ^2 , called chi-square.

TEST STATISTIC A value, determined from sample information, used to determine whether to reject the null hypothesis.

In hypothesis testing for the mean (μ) when σ is known, the test statistic z is computed by:

TESTING A MEAN, σ KNOWN

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

(10–1)

The z value is based on the sampling distribution of $\bar{x}$, which follows the normal distribution with a mean ($\mu_{\bar{x}}$) equal to μ and a standard deviation $\sigma_{\bar{x}}$, which is equal to $\sigma/\sqrt{n}$. We can thus determine whether the difference between $\bar{x}$ and μ is statistically significant by finding the number of standard deviations $\bar{x}$ is from μ , using formula (10–1).

Step 4: Formulate the Decision Rule

A decision rule is a statement of the specific conditions under which the null hypothesis is rejected and the conditions under which it is not rejected. The region or area of rejection defines the location of all those values that are so large or so small that the probability of their occurrence under a true null hypothesis is rather remote.

Chart 10–1 portrays the rejection region for a test of significance that will be conducted later in the chapter.

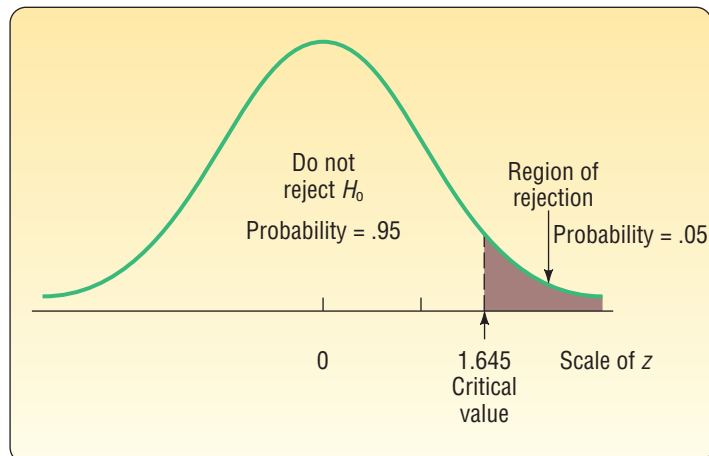


CHART 10–1 Sampling Distribution of the Statistic z , a Right-Tailed Test, .05 Level of Significance

STATISTICS IN ACTION

During World War II, allied military planners needed estimates of the number of German tanks. The information provided by traditional spying methods was not reliable, but statistical methods proved to be valuable. For example, espionage and reconnaissance led analysts to estimate that 1,550 tanks were produced during June 1941. However, using the serial numbers of captured tanks and statistical analysis, military planners estimated that only 244 tanks were produced. The actual number produced, as determined from German production records, was 271. The estimate using statistical analysis turned out to be much more accurate. A similar type of analysis was used to estimate the number of Iraqi tanks destroyed during Desert Storm.

Note in the chart that:

- The area where the null hypothesis is not rejected is to the left of 1.645. We will explain how to get the 1.645 value shortly.
- The area of rejection is to the right of 1.645.
- A one-tailed test is being applied. (This will also be explained later.)
- The .05 level of significance was chosen.
- The sampling distribution of the statistic z follows the normal probability distribution.
- The value 1.645 separates the regions where the null hypothesis is rejected and where it is not rejected.
- The value 1.645 is the **critical value**.

CRITICAL VALUE The dividing point between the region where the null hypothesis is rejected and the region where it is not rejected.

Step 5: Make a Decision

The fifth step in hypothesis testing is to compute the value of the test statistic, compare its value to the critical value, and make a decision to reject or not to reject the null hypothesis. Referring to Chart 10–1, if, based on sample information, z is computed to be 2.34, the null hypothesis is rejected at the .05 level of significance. The decision to reject H_0 was made because 2.34 lies in the region of rejection, that is, beyond 1.645. We reject the null hypothesis, reasoning that it is highly improbable that a computed z value this large is due to sampling error (chance).

Had the computed value been 1.645 or less, say 0.71, the null hypothesis is not rejected. It is reasoned that such a small computed value could be attributed to chance, that is, sampling error. As we have emphasized, only one of two decisions is possible in hypothesis testing—either reject or do not reject the null hypothesis.

However, because the decision is based on a sample, it is always possible to make either of two decision errors. It is possible to make a Type I error when the null hypothesis is rejected when it should not be rejected. Or it is also possible to make a Type II error when the null hypothesis is not rejected and it should have been rejected. Fortunately, we select the probability of making a Type I error, α (alpha), and we can compute the probabilities associated with a Type II error, β (beta).

Step 6: Interpret the Result

The final step in the hypothesis testing procedure is to interpret the results. The process does not end with the value of a sample statistic or the decision to reject or not reject the null hypothesis. What can we say or report based on the results of the statistical test? Here are two examples:

- An investigative reporter for a Colorado newspaper reports that the mean monthly income of convenience stores in the state is \$130,000. You decide to conduct a test of hypothesis to verify the report. The null hypothesis and the alternate hypothesis are:

$$H_0: \mu = \$130,000$$

$$H_1: \mu \neq \$130,000$$

A sample of convenience stores provides a sample mean and standard deviation, and you compute a z statistic. The results of the hypothesis test result in a decision to not reject the null hypothesis. How do you interpret the result? Be cautious with

STATISTICS IN ACTION

LASIK is a 15-minute surgical procedure that uses a laser to reshape an eye's cornea with the goal of improving eyesight. Research shows that about 5% of all surgeries involve complications such as glare, corneal haze, overcorrection or undercorrection of vision, and loss of vision. In a statistical sense, the research tests a null hypothesis that the surgery will not improve eyesight with the alternative hypothesis that the surgery will improve eyesight. The sample data of LASIK surgery shows that 5% of all cases result in complications. The 5% represents a Type I error rate. When a person decides to have the surgery, he or she expects to reject the null hypothesis. In 5% of future cases, this expectation will not be met. (Source: *American Academy of Ophthalmology Journal*, Vol. 16, no. 43.)

your interpretation because by not rejecting the null hypothesis, you did not prove the null hypothesis to be true. Based on the sample data, the difference between the sample mean and hypothesized population mean was not large enough to reject the null hypothesis.

- In a recent speech to students, the dean of the College of Business reported that the mean credit card debt for college students is \$3,000. You decide to conduct a test of the dean's statement or hypothesis to investigate the statement's truth. The null hypothesis and the alternate hypothesis are:

$$H_0: \mu = \$3,000$$

$$H_1: \mu \neq \$3,000$$

A random sample of college students provides a sample mean and standard deviation, and you compute a z statistic. The hypothesis test results in a decision to reject the null hypothesis. How do you interpret the result? The sample evidence does not support the dean's statement. Based on the sample data, the mean amount of student credit card debt is different from \$3,000. You have disproved the null hypothesis with a stated probability of a Type I error, α . That is, there is a small probability that the decision to reject the null hypothesis was an error due to random sampling.

SUMMARY OF THE STEPS IN HYPOTHESIS TESTING

1. Establish the null hypothesis (H_0) and the alternate hypothesis (H_1).
2. Select the level of significance, that is, α .
3. Select an appropriate test statistic.
4. Formulate a decision rule based on steps 1, 2, and 3 above.
5. Make a decision regarding the null hypothesis based on the sample information.
6. Interpret the results of the test.

Before actually conducting a test of hypothesis, we describe the difference between a one-tailed and a two-tailed hypothesis test.

LO10-3

Distinguish between a one-tailed and a two-tailed test of hypothesis.

ONE-TAILED AND TWO-TAILED HYPOTHESIS TESTS

Refer to Chart 10–1. It shows a one-tailed test. It is called a one-tailed test because the rejection region is only in one tail of the curve. In this case, it is in the right, or upper, tail of the curve. To illustrate, suppose that the packaging department at General Foods Corporation is concerned that some boxes of Grape Nuts are significantly overweight. The cereal is packaged in 453-gram boxes, so the null hypothesis is $H_0: \mu \leq 453$. This is read, “the population mean (μ) is equal to or less than 453.” The alternate hypothesis is, therefore, $H_1: \mu > 453$. This is read, “ μ is greater than 453.” Note that the inequality sign in the alternate hypothesis ($>$) points to the region of rejection in the upper tail. (See Chart 10–1.) Also observe that the null hypothesis includes the equal sign. That is, $H_0: \mu \leq 453$. The equality condition always appears in H_0 , never in H_1 .

Chart 10–2 portrays a situation where the rejection region is in the left (lower) tail of the standard normal distribution. As an illustration, consider the problem of automobile manufacturers, large automobile leasing companies, and other organizations that purchase large quantities of tires. They want the tires to average, say, 60,000 miles of wear under normal usage. They will, therefore, reject a shipment of tires if tests reveal that the mean life of the tires is significantly below 60,000 miles. They gladly accept a shipment if the mean life is greater than 60,000 miles! They are not concerned with this possibility, however. They are concerned only if they have sample evidence to conclude

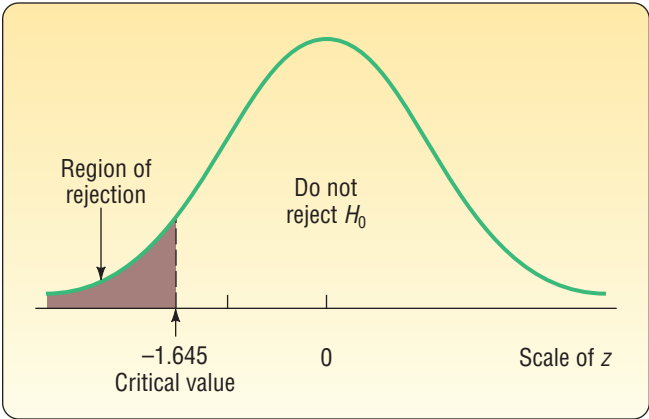


CHART 10–2 Sampling Distribution for the Statistic z , Left-Tailed Test, .05 Level of Significance

that the tires will average less than 60,000 miles of useful life. Thus, the test is set up to satisfy the concern of the automobile manufacturers that *the mean life of the tires is not less than 60,000 miles*. This statement appears in the null hypothesis. The null and alternate hypotheses in this case are written $H_0: \mu \geq 60,000$ and $H_1: \mu < 60,000$.

One way to determine the location of the rejection region is to look at the direction in which the inequality sign in the alternate hypothesis is pointing (either $<$ or $>$). In the tire wear problem, it is pointing to the left, and the rejection region is therefore in the left tail.

In summary, a test is *one-tailed* when the alternate hypothesis, H_1 , states a direction, such as:

- H_0 : The mean income of female stockbrokers is *less than or equal to* \$65,000 per year.
- H_1 : The mean income of female stockbrokers is *greater than* \$65,000 per year.

If no direction is specified in the alternate hypothesis, we use a *two-tailed* test. Changing the previous problem to illustrate, we can say:

- H_0 : The mean income of female stockbrokers is \$65,000 per year.
- H_1 : The mean income of female stockbrokers is *not equal to* \$65,000 per year.

If the null hypothesis is rejected and H_1 accepted in the two-tailed case, the mean income could be significantly greater than \$65,000 per year or it could be significantly less than \$65,000 per year. To accommodate these two possibilities, the 5% area of rejection is divided equally into the two tails of the sampling distribution (2.5% each). Chart 10–3 shows the two areas and the critical values. Note that the total area in the normal distribution is 1.0000, found by $.9500 + .0250 + .0250$.

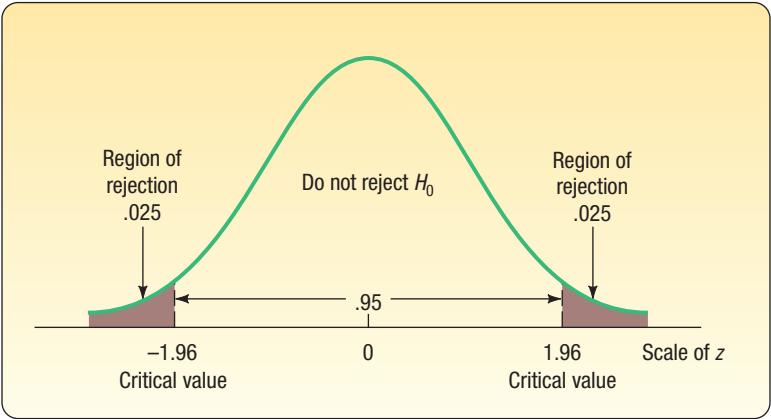


CHART 10–3 Regions of Nonrejection and Rejection for a Two-Tailed Test, .05 Level of Significance

LO10-4

Conduct a test of a hypothesis about a population mean.

HYPOTHESIS TESTING FOR A POPULATION MEAN: KNOWN POPULATION STANDARD DEVIATION

A Two-Tailed Test

An example will show the details of the six-step hypothesis testing procedure. We also wish to use a two-tailed test. That is, we are *not* concerned whether the sample results are larger or smaller than the proposed population mean. Rather, we are interested in whether it is *different from* the proposed value for the population mean. We begin, as we did in the previous chapter, with a situation in which we have historical information about the population and in fact know its standard deviation.

EXAMPLE

Jamestown Steel Company manufactures and assembles desks and other office equipment at several plants in western New York State. The weekly production of the Model A325 desk at the Fredonia Plant follows a normal probability distribution with a mean of 200 and a standard deviation of 16. Recently, because of market expansion, new production methods have been introduced and new employees hired. The vice president of manufacturing would like to investigate whether there has been a *change* in the weekly production of the Model A325 desk. Is the mean number of desks produced at the Fredonia Plant *different from* 200 at the .01 significance level?



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SOLUTION

In this example, we know two important pieces of information: (1) the population of weekly production follows the normal distribution and (2) the standard deviation of this normal distribution is 16 desks per week. So it is appropriate to use the *z* statistic. We use the statistical hypothesis testing procedure to investigate whether the production rate has changed from 200 per week.

Step 1: State the null hypothesis and the alternate hypothesis. The null hypothesis is “The population mean is 200.” The alternate hypothesis is “The mean is different from 200” or “The mean is not 200.” These two hypotheses are written:

$$H_0: \mu = 200$$

$$H_1: \mu \neq 200$$

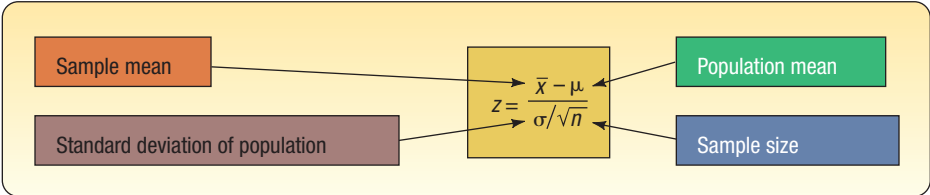
This is a *two-tailed test* because the alternate hypothesis does not state a direction. In other words, it does not state whether the mean production is greater than 200 or less than 200. The vice president wants only to find out whether the production rate is different from 200.

Before moving to Step 2, we wish to emphasize two points.

- The null hypothesis has the equal sign. Why? Because the value we are testing is always in the null hypothesis. Logically, the alternate hypothesis never contains the equal sign.
- Both the null hypothesis and the alternate hypothesis contain Greek letters—in this case μ , which is the symbol for the population mean. Tests of hypothesis **always** refer to population parameters, never to sample statistics. To put it another way, you will never see the symbol $\bar{x}$ as part of the null hypothesis or the alternate hypothesis.

Step 2: Select the level of significance. In the example description, the significance level selected is .01. This is α , the probability of committing a Type I error, and it is the probability of rejecting a true null hypothesis.

Step 3: Select the test statistic. The test statistic is z when the population standard deviation is known. Transforming the production data to standard units (z values) permits their use not only in this problem but also in other hypothesis-testing problems. Formula (10–1) for z is repeated next with the various letters identified.



Step 4: Formulate the decision rule. We formulate the decision rule by first determining the critical values of z . Because this is a two-tailed test, half of .01, or .005, is placed in each tail. The area where H_0 is not rejected, located between the two tails, is therefore .99. Using the Student’s t Distribution table in Appendix B.5, move to the top margin called “Level of Significance for Two-Tailed Tests, α ,” select the column with $\alpha = .01$, and move to the last row, which is labeled ∞ , or infinite degrees of freedom. The z value in this cell is 2.576. For your convenience, Appendix B.5, Student’s t Distribution, is repeated in the inside back cover. All the facets of this problem are shown in the diagram in Chart 10–4.

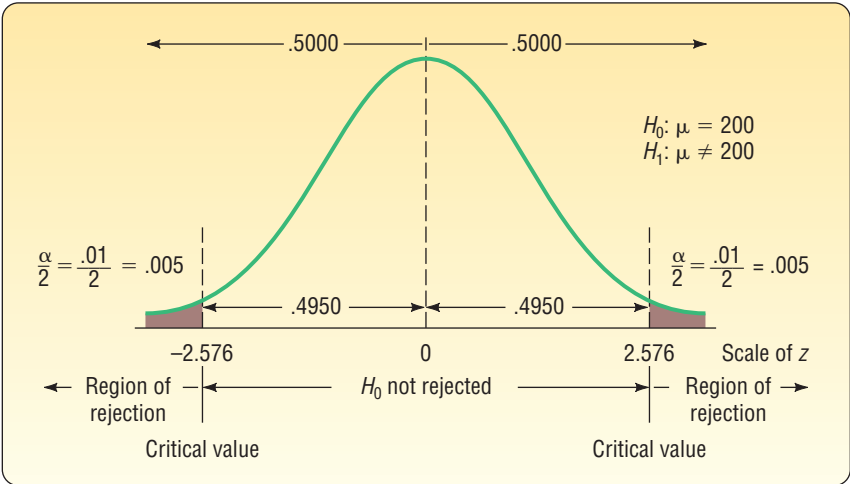


CHART 10–4 Decision Rule for the .01 Significance Level

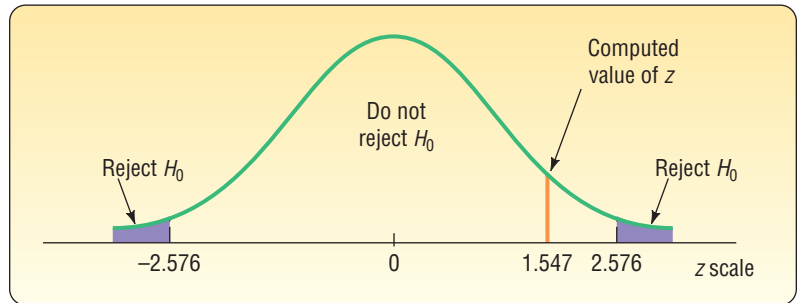
The decision rule is: if the computed value of z is not between -2.576 and 2.576 , reject the null hypothesis. If z falls between -2.576 and 2.576 , do not reject the null hypothesis.

Step 5: Make a decision. Take a sample from the population (weekly production), compute a test statistic, apply the decision rule, and arrive at a decision to reject H_0 or not to reject H_0 . The mean number of desks produced last year (50 weeks because the plant was shut down 2 weeks for vacation) is 203.5. The standard deviation of the population is 16 desks per week. Computing the z value from formula (10–1):

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{203.5 - 200}{16/\sqrt{50}} = 1.547$$

Because 1.547 is between -2.576 and 2.576 , we decide not to reject H_0 .

Step 6: Interpret the result. We did not reject the null hypothesis, so we have failed to show that the population mean has changed from 200 per week. To put it another way, the difference between the population mean of 200 per week and the sample mean of 203.5 could simply be due to chance. What should we tell the vice president? The sample information fails to indicate that the new production methods resulted in a change in the 200-desks-per-week production rate.



Did we prove that the assembly rate is still 200 per week? Not really. *We failed to disprove the null hypothesis.* Failing to disprove the hypothesis that the population mean is 200 is not the same thing as proving it to be true. For example, in the U.S. judicial system, a person is presumed innocent until proven guilty. The trial starts with a null hypothesis that the individual is innocent. If the individual is acquitted, the trial did not provide enough evidence to reject the presumption of innocence and conclude that the individual was not innocent or guilty as charged. That is what we do in statistical hypothesis testing when we do not reject the null hypothesis. The correct interpretation is that, based on the evidence or sample information, we have failed to disprove the null hypothesis.

We selected the significance level, .01 in this case, before setting up the decision rule and sampling the population. This is the appropriate strategy. The significance level should be set by the investigator, but it should be determined *before* gathering the sample evidence and not changed based on the sample evidence.

How does the hypothesis testing procedure just described compare with that of confidence intervals discussed in the previous chapter? When we conducted the test of hypothesis regarding the production of desks, we changed the units from desks per

week to a z value. Then we compared the computed value of the test statistic (1.547) to that of the critical values (-2.576 and 2.576). Because the computed value of the test statistic was in the region where the null hypothesis was not rejected, we concluded that the population mean could be 200. To use the confidence interval approach, on the other hand, we would develop a confidence interval, based on formula (9-1). See page 287. The interval would be from 197.671 to 209.329, found by $203.5 \pm 2.576(16/\sqrt{50})$. Note that the proposed population value, 200, is within this interval. Hence, we would conclude that the population mean could reasonably be 200.

In general, H_0 is rejected if the confidence interval does not include the hypothesized value. If the confidence interval includes the hypothesized value, then H_0 is not rejected. So the “do not reject region” for a test of hypothesis is equivalent to the proposed population value occurring in the confidence interval.

SELF-REVIEW 10-1



Heinz, a manufacturer of ketchup, uses a particular machine to dispense 16 ounces of its ketchup into containers. From many years of experience with the particular dispensing machine, Heinz knows the amount of product in each container follows a normal distribution with a mean of 16 ounces and a standard deviation of 0.15 ounce. A sample of 50 containers filled last hour revealed the mean amount per container was 16.017 ounces. Does this evidence suggest that the mean amount dispensed is different from 16 ounces? Use the .05 significance level.



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- State the null hypothesis and the alternate hypothesis.
- What is the probability of a Type I error?
- Give the formula for the test statistic.
- State the decision rule.
- Determine the value of the test statistic.
- What is your decision regarding the null hypothesis?
- Interpret, in a single sentence, the result of the statistical test.

A One-Tailed Test

In the previous example/solution, we emphasized that we were concerned only with reporting to the vice president whether there had been a change in the mean number of desks assembled at the Fredonia Plant. We were not concerned with whether the change was an increase or a decrease in the production.

To illustrate a one-tailed test, let's change the problem. Suppose the vice president wants to know whether there has been an *increase* in the number of units assembled. Can we conclude, because of the improved production methods, that the mean number of desks assembled in the last 50 weeks was more than 200? Look at the difference in the way the problem is formulated. In the first case, we wanted to know whether there was a *difference* in the mean number assembled, but now we want to know whether there has been an *increase*. Because we are investigating different questions, we will set our hypotheses differently. The biggest difference occurs in the alternate hypothesis. Before, we stated the alternate hypothesis as “different from”; now we want to state it as “greater than.” In symbols:

A two-tailed test:

$$H_0: \mu = 200$$

$$H_1: \mu \neq 200$$

A one-tailed test:

$$H_0: \mu \leq 200$$

$$H_1: \mu > 200$$

The critical values for a one-tailed test are different from a two-tailed test at the same significance level. In the previous example/solution, we split the significance level in half and put half in the lower tail and half in the upper tail. In a one-tailed test, we put all the rejection region in one tail. See Chart 10–5.

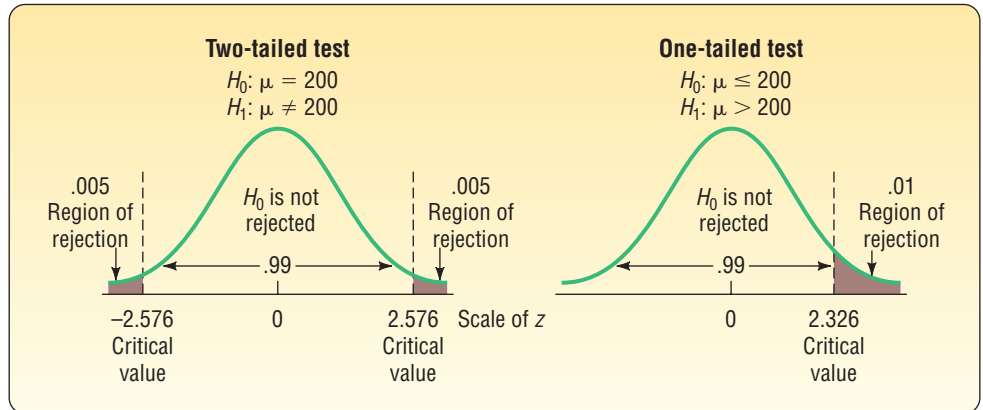


CHART 10–5 Rejection Regions for Two-Tailed and One-Tailed Tests, $\alpha = .01$

For the one-tailed test, the critical value of z is 2.326. Using the Student's t Distribution table in Appendix B.5, move to the top heading called “Level of Significance for One-Tailed Tests, α ,” select the column with $\alpha = .01$, and move to the last row, which is labeled ∞ , or infinite degrees of freedom. The z value in this cell is 2.326.

LO10-5

Compute and interpret a p -value.

p -VALUE IN HYPOTHESIS TESTING

In testing a hypothesis, we compare the test statistic to a critical value. A decision is made to either reject or not reject the null hypothesis. So, for example, if the critical value is 1.96 and the computed value of the test statistic is 2.19, the decision is to reject the null hypothesis.

In recent years, spurred by the availability of computer software, additional information is often reported on the strength of the rejection. That is, how confident are we in rejecting the null hypothesis? This approach reports the probability (assuming that the null hypothesis is true) of getting a value of the test statistic at least as extreme as the value actually obtained. This process compares the probability, called the **p -value**, with the significance level. If the p -value is smaller than the significance level, H_0 is rejected. If it is larger than the significance level, H_0 is not rejected.

STATISTICS IN ACTION

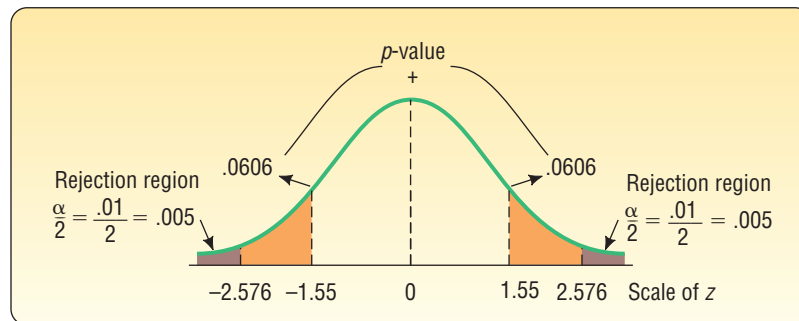
There is a difference between *statistically significant* and *practically significant*. To explain, suppose we develop a new diet pill and test it on 100,000 people. We conclude that the typical person taking the pill for 2 years lost 1 pound. Do you think many people would be interested in taking the pill to lose 1 pound? The results of using the new pill were statistically significant but not practically significant.

p -VALUE The probability of observing a sample value as extreme as, or more extreme than, the value observed, given that the null hypothesis is true.

Determining the p -value not only results in a decision regarding H_0 , but it gives us additional insight into the strength of the decision. A very small p -value, such as .0001, indicates that there is little likelihood the H_0 is true. On the other hand, a p -value of .2033 means that H_0 is not rejected, and there is little likelihood that it is false.

How do we find the p -value? To calculate p -values, we will need to use the z table (Appendix B.3) and, to use this table, we will round z test statistics to two decimals. To illustrate how to compute a p -value, we will use the example where we tested the null hypothesis that the mean number of desks produced per week at Fredonia was 200.

We did not reject the null hypothesis because the computed z test statistic of 1.547 fell in the region between -2.576 and 2.576 . We agreed not to reject the null hypothesis if the z test statistic fell in this region. Rounding 1.547 to 1.55 and using the z table, the probability of finding a z value of 1.55 or more is .0606, found by $.5000 - .4394$. To put it another way, the probability of obtaining an $\bar{x}$ greater than 203.5 if $\mu = 200$ is .0606. To compute the p -value, we need to be concerned with the region less than -1.55 as well as the values greater than 1.55 (because the rejection region is in both tails). The two-tailed p -value is .1212, found by $2(.0606)$. The p -value of .1212 is greater than the significance level of .01 decided upon initially, so H_0 is not rejected. The details are shown in the following graph. Notice for the two-tailed hypothesis test, the p -value is represented by areas in both tails of the distribution. Then the p -value can easily be compared with the significance level. The same decision rule is used as in the one-sided test.



A p -value is a way to express the likelihood that H_0 is false. But how do we interpret a p -value? We have already said that if the p -value is less than the significance level, then we reject H_0 ; if it is greater than the significance level, then we do not reject H_0 . Also, if the p -value is very large, then it is likely that H_0 is true. If the p -value is small, then it is likely that H_0 is not true. The following box will help to interpret p -values.

INTERPRETING THE WEIGHT OF EVIDENCE AGAINST H_0

If the p -value is less than

- (a) .10, we have *some* evidence that H_0 is not true.
- (b) .05, we have *strong* evidence that H_0 is not true.
- (c) .01, we have *very strong* evidence that H_0 is not true.
- (d) .001, we have *extremely strong* evidence that H_0 is not true.

SELF-REVIEW 10-2



Refer to Self-Review 10-1.

- (a) Suppose the next to the last sentence is changed to read: Does this evidence suggest that the mean amount dispensed is *more than* 16 ounces? State the null hypothesis and the alternate hypothesis under these conditions.
- (b) What is the decision rule under the new conditions stated in part (a)?
- (c) A second sample of 50 filled containers revealed the mean to be 16.040 ounces. What is the value of the test statistic for this sample?
- (d) What is your decision regarding the null hypothesis?
- (e) Interpret, in a single sentence, the result of the statistical test.
- (f) What is the p -value? What is your decision regarding the null hypothesis based on the p -value? Is this the same conclusion reached in part (d)?

EXERCISES

For Exercises 1–4, answer the questions: (a) Is this a one- or two-tailed test? (b) What is the decision rule? (c) What is the value of the test statistic? (d) What is your decision regarding H_0 ? (e) What is the p -value? Interpret it.

1. A sample of 36 observations is selected from a normal population. The sample mean is 49, and the population standard deviation is 5. Conduct the following test of hypothesis using the .05 significance level.

$$H_0: \mu = 50$$

$$H_1: \mu \neq 50$$

2. A sample of 36 observations is selected from a normal population. The sample mean is 12, and the population standard deviation is 3. Conduct the following test of hypothesis using the .01 significance level.

$$H_0: \mu \leq 10$$

$$H_1: \mu > 10$$

3. A sample of 36 observations is selected from a normal population. The sample mean is 21, and the population standard deviation is 5. Conduct the following test of hypothesis using the .05 significance level.

$$H_0: \mu \leq 20$$

$$H_1: \mu > 20$$

4. A sample of 64 observations is selected from a normal population. The sample mean is 215, and the population standard deviation is 15. Conduct the following test of hypothesis using the .025 significance level.

$$H_0: \mu \geq 220$$

$$H_1: \mu < 220$$

For Exercises 5–8: (a) State the null hypothesis and the alternate hypothesis. (b) State the decision rule. (c) Compute the value of the test statistic. (d) What is your decision regarding H_0 ? (e) What is the p -value? Interpret it.

5. The manufacturer of the X-15 steel-belted radial truck tire claims that the mean mileage the tire can be driven before the tread wears out is 60,000 miles. Assume the mileage wear follows the normal distribution and the standard deviation of the distribution is 5,000 miles. Crosset Truck Company bought 48 tires and found that the mean mileage for its trucks is 59,500 miles. Is Crosset's experience different from that claimed by the manufacturer at the .05 significance level?
6. The waiting time for customers at MacBurger Restaurants follows a normal distribution with a population standard deviation of 1 minute. At the Warren Road MacBurger, the quality-assurance department sampled 50 customers and found that the mean waiting time was 2.75 minutes. At the .05 significance level, can we conclude that the mean waiting time is less than 3 minutes?
7. A recent national survey found that high school students watched an average (mean) of 6.8 movies per month with a population standard deviation of 1.8. The distribution of number of movies watched per month follows the normal distribution. A random sample of 36 college students revealed that the mean number of movies watched last month was 6.2. At the .05 significance level, can we conclude that college students watch fewer movies a month than high school students?
8. At the time she was hired as a server at the Grumney Family Restaurant, Beth Brigden was told, "You can average \$80 a day in tips." Assume the population of daily tips is normally distributed with a standard deviation of \$9.95. Over the first 35 days she was employed at the restaurant, the mean daily amount of her tips was \$84.85. At the .01 significance level, can Ms. Brigden conclude that her daily tips average more than \$80?

LO10-6

Use a *t* statistic to test a hypothesis.

HYPOTHESIS TESTING FOR A POPULATION MEAN: POPULATION STANDARD DEVIATION UNKNOWN

In the preceding example, we knew σ , the population standard deviation, and that the population followed the normal distribution. In most cases, however, the population standard deviation is unknown. Thus, σ must be based on prior studies or estimated by the sample standard deviation, *s*. The population standard deviation in the following example is not known, so the sample standard deviation is used to estimate σ .

To find the value of the test statistic, we use the *t* distribution and revise formula (10–1) as follows:

TESTING A MEAN, σ UNKNOWN

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

(10–2)

with *n* – 1 degrees of freedom, where:

- $\bar{x}$ is the sample mean.
- μ is the hypothesized population mean.
- s* is the sample standard deviation.
- n* is the number of observations in the sample.

We encountered this same situation when constructing confidence intervals in the previous chapter. See pages 292–294 in Chapter 9. We summarized this problem in Chart 9–3 on page 294. Under these conditions, the correct statistical procedure is to replace the standard normal distribution with the *t* distribution. To review, the major characteristics of the *t* distribution are:

- It is a continuous distribution.
- It is bell-shaped and symmetrical.
- There is a family of *t* distributions. Each time the degrees of freedom change, a new distribution is created.
- As the number of degrees of freedom increases, the shape of the *t* distribution approaches that of the standard normal distribution.
- The *t* distribution is flatter, or more spread out, than the standard normal distribution.

The following example/solution shows the details.

EXAMPLE

The McFarland Insurance Company Claims Department reports the mean cost to process a claim is \$60. An industry comparison showed this amount to be larger than most other insurance companies, so the company instituted cost-cutting measures. To evaluate the effect of the cost-cutting measures, the supervisor of the Claims Department selected a random sample of 26 claims processed last month and recorded the cost to process each claim. The sample information is reported below.

\$45	\$49	\$62	\$40	\$43	\$61
48	53	67	63	78	64
48	54	51	56	63	69
58	51	58	59	56	57
38	76				

At the .01 significance level, is it reasonable to conclude that the mean cost to process a claim is now less than \$60?

SOLUTION

We will use the six-step hypothesis testing procedure.

Step 1: State the null hypothesis and the alternate hypothesis. The null hypothesis is that the population mean is at least \$60. The alternate hypothesis is that the population mean is less than \$60. We can express the null and alternate hypotheses as follows:

$$H_0: \mu \geq \$60$$

$$H_1: \mu < \$60$$

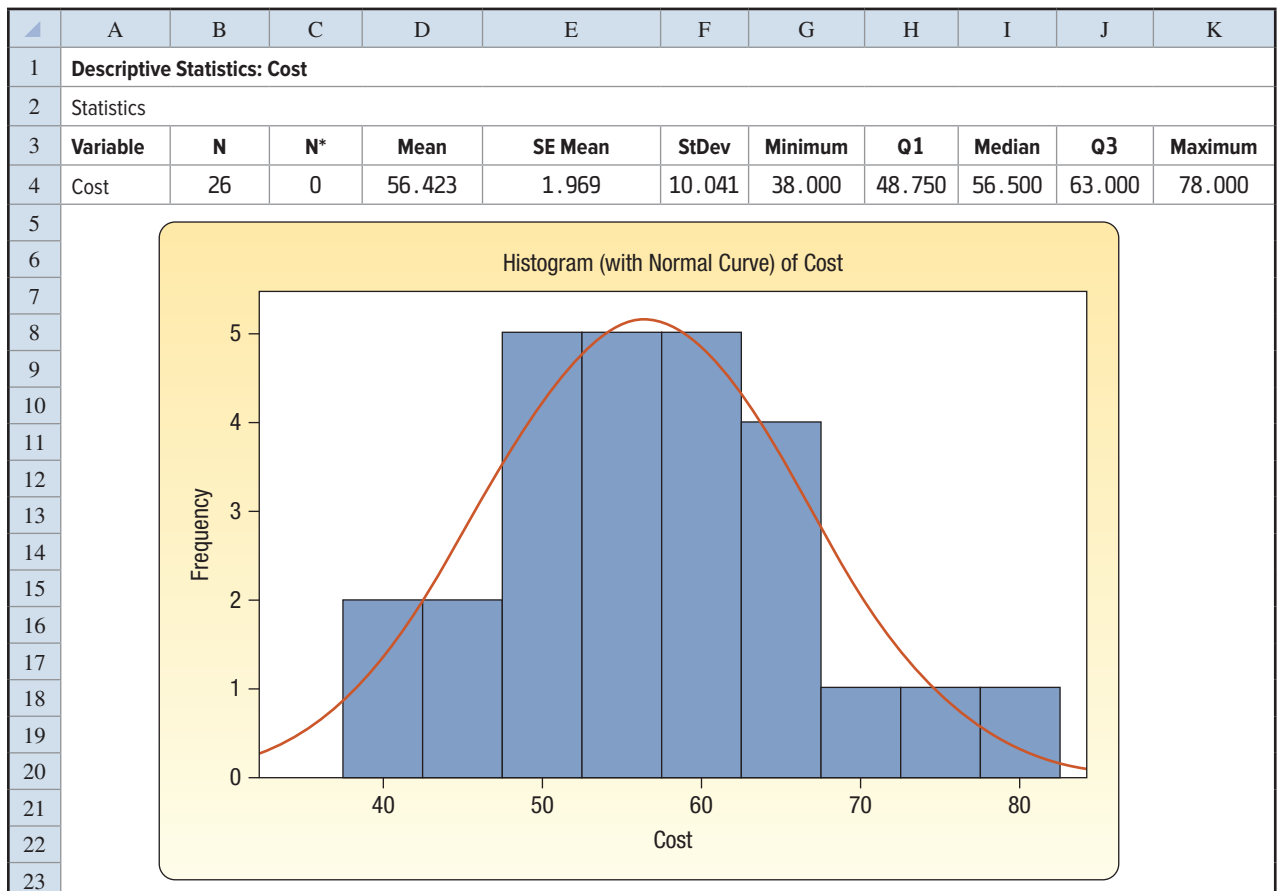
The test is *one-tailed* because we want to determine whether there has been a *reduction* in the cost. The inequality in the alternate hypothesis points to the region of rejection in the left tail of the distribution.

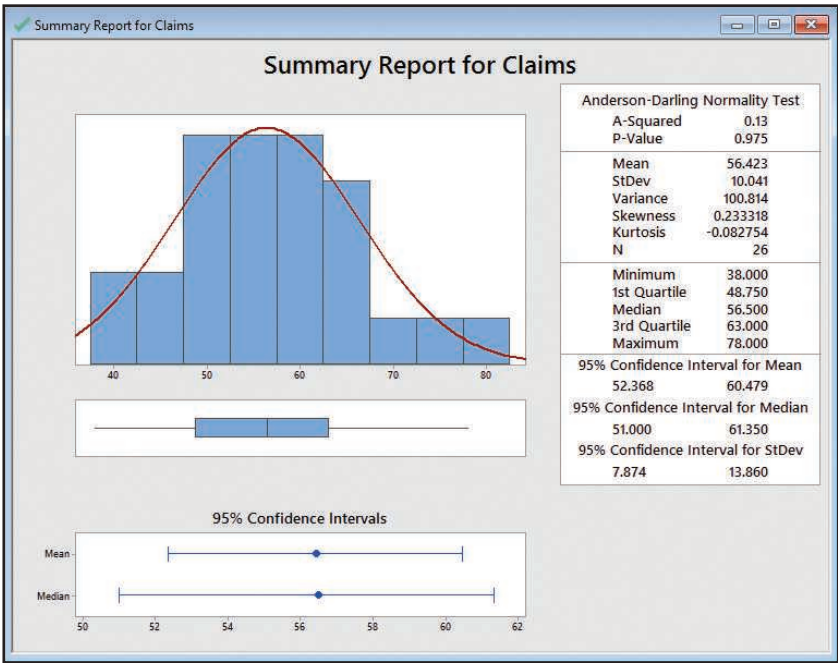
Step 2: Select the level of significance. We decided on the .01 significance level.

Step 3: Select the test statistic. The test statistic in this situation is the *t* distribution. Why? First, it is reasonable to conclude that the distribution of the cost per claim follows the normal distribution. We can confirm this from the histogram in the center of the following Minitab output. Observe the normal distribution superimposed on the frequency distribution.

We do not know the standard deviation of the population. So we substitute the sample standard deviation. The value of the test statistic is computed by formula (10–2):

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$





Step 4: Formulate the decision rule. The critical values of t are given in Appendix B.5, a portion of which is shown in Table 10–1. Appendix B.5 is also repeated in the back inside cover of the text. The far left column of the table is labeled “ df ” for degrees of freedom. The number of degrees of freedom is the total number of observations in the sample minus the number of populations sampled, written $n - 1$. In this case, the number of observations in the sample is 26, and we sampled 1 population, so there are $26 - 1 = 25$ degrees of freedom.

TABLE 10–1 A Portion of the t Distribution Table

Confidence Intervals						
	80%	90%	95%	98%	99%	99.9%
df	Level of Significance for One-Tailed Test, α					
	0.10	0.05	0.025	0.01	0.005	0.0005
	Level of Significance for Two-Tailed Test, α					
	0.20	0.10	0.05	0.02	0.01	0.001
⋮	⋮	⋮	⋮	⋮	⋮	⋮
21	1.323	1.721	2.080	2.518	2.831	3.819
22	1.321	1.717	2.074	2.508	2.819	3.792
23	1.319	1.714	2.069	2.500	2.807	3.768
24	1.318	1.711	2.064	2.492	2.797	3.745
25	1.316	1.708	2.060	2.485	2.787	3.725
26	1.315	1.706	2.056	2.479	2.779	3.707
27	1.314	1.703	2.052	2.473	2.771	3.690
28	1.313	1.701	2.048	2.467	2.763	3.674
29	1.311	1.699	2.045	2.462	2.756	3.659
30	1.310	1.697	2.042	2.457	2.750	3.646

To find the critical value, first locate the row with the appropriate degrees of freedom. This row is shaded in Table 10–1. Next, determine whether the test is one-tailed or two-tailed. In this case, we have a one-tailed test, so find the portion of the table that is labeled “one-tailed.” Locate the column with the selected significance level. In this example, the significance level is .01. Move down the column labeled “0.01” until it intersects the row with 25 degrees of freedom. The value is 2.485. Because this is a one-sided test and the rejection region is in the left tail, the critical value is negative. The decision rule is to reject H_0 if the value of t is less than -2.485 .

Step 5: Make a decision. From the Minitab output, the mean cost per claim for the sample of 26 observations is \$56.423. The standard deviation of this sample is \$10.041. We insert these values in formula (10–2) and compute the value of t :

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{\$56.423 - \$60}{\$10.041/\sqrt{26}} = -1.816$$

Because -1.816 lies in the region to the right of the critical value of -2.485 (see Chart 10–6), the null hypothesis is not rejected at the .01 significance level.

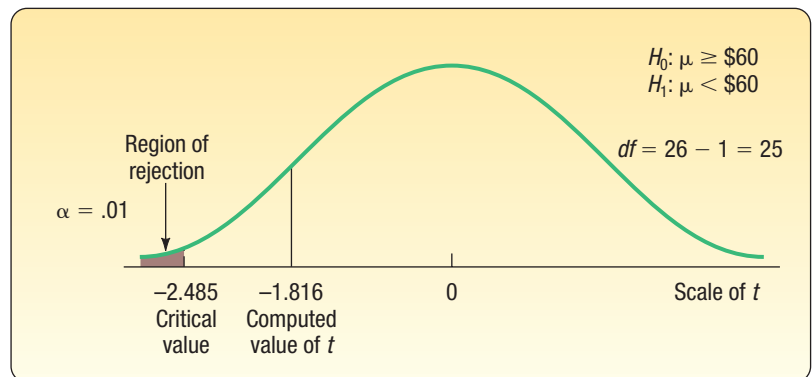


CHART 10–6 Rejection Region, t Distribution, .01 Significance Level

Step 6: Interpret the result. We have not disproved the null hypothesis. The sample of claims could have been selected from a population with a mean cost of \$60 per claim. To put it another way, the difference of \$3.577 (\$56.423 – \$60.00) between the sample mean and the population mean could be due to sampling error. The test results do not allow the claims department manager to conclude that the cost-cutting measures have been effective.

In the previous example, the mean and the standard deviation were computed using Minitab. The following example/solution shows the details when the sample mean and sample standard deviation are calculated from sample data.

EXAMPLE

The Myrtle Beach International Airport provides a cell phone parking lot where people can wait for a message to pick up arriving passengers. To decide if the cell phone lot has enough parking places, the manager of airport parking needs to

know if the mean time in the lot is more than 15 minutes. A sample of 12 recent customers showed they were in the lot the following lengths of time, in minutes.

30	24	28	22	14	2	39	23	23	28	12	31
----	----	----	----	----	---	----	----	----	----	----	----

At the .05 significance level, is it reasonable to conclude that the mean time in the lot is more than 15 minutes?

SOLUTION

We begin by stating the null hypothesis and the alternate hypothesis. In this case, the question is whether the population mean could be more than 15 minutes. So this is a one-tailed test. We state the two hypotheses as follows:

$H_0: \mu \leq 15$

$H_1: \mu > 15$

There are 11 degrees of freedom, found by $n - 1 = 12 - 1 = 11$. The critical t value is 1.796, found by referring to Appendix B.5 for a one-tailed test, using $\alpha = .05$ with 11 degrees of freedom. The decision rule is: Reject the null hypothesis if the computed t is greater than 1.796. This information is summarized in Chart 10–7.

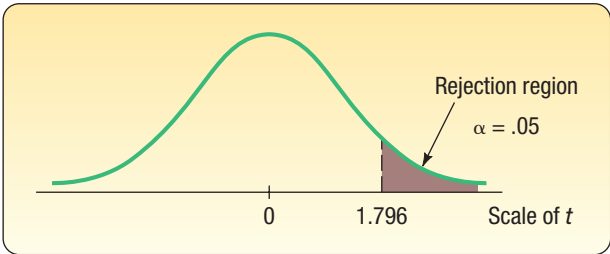


CHART 10–7 Rejection Region, One-Tailed Test, Student’s t Distribution, $\alpha = .05$

We calculate the sample mean using formula (3–2) and the sample standard deviation using formula (3–8). The sample mean is 23 minutes, and the sample standard deviation is 9.835 minutes. The details of the calculations are shown in Table 10–2.

TABLE 10–2 Calculations of Sample Mean and Standard Deviation Parking Times

Customer	x , Minutes	$(x - \bar{x})^2$
Chmura	30	49
Will	24	1
Crompton	28	25
Craver	22	1
Cao	14	81
Nowlin	2	441
Esposito	39	256
Colvard	23	0
Hoefle	23	0
Lawler	28	25
Trask	12	121
Grullon	31	64
Total	276	1064

$$\bar{x} = \frac{\Sigma x}{n} = \frac{276}{12} = 23$$
$$s = \sqrt{\frac{\Sigma (x - \bar{x})^2}{n - 1}} = \sqrt{\frac{1064}{12 - 1}} = 9.835$$

Now we are ready to compute the value of t , using formula (10-2).

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{23 - 15}{9.835/\sqrt{12}} = 2.818$$

The null hypothesis that the population mean is less than or equal to 15 minutes is rejected because the computed t value of 2.818 lies in the area to the right of 1.796. We conclude that the time customers spend in the lot is more than 15 minutes. This result indicates that the airport may need to add more parking places.

SELF-REVIEW 10-3



The mean life of a battery used in a digital clock is 305 days. The lives of the batteries follow the normal distribution. The battery was recently modified to last longer. A sample of 20 of the modified batteries had a mean life of 311 days with a standard deviation of 12 days. Did the modification increase the mean life of the battery?

- State the null hypothesis and the alternate hypothesis.
- Show the decision rule graphically. Use the .05 significance level.
- Compute the value of t . What is your decision regarding the null hypothesis? Briefly summarize your results.

EXERCISES

9. Given the following hypotheses:

$$H_0: \mu \leq 10$$

$$H_1: \mu > 10$$

A random sample of 10 observations is selected from a normal population. The sample mean was 12 and the sample standard deviation 3. Using the .05 significance level:

- State the decision rule.
- Compute the value of the test statistic.
- What is your decision regarding the null hypothesis?

10. Given the following hypotheses:

$$H_0: \mu = 400$$

$$H_1: \mu \neq 400$$

A random sample of 12 observations is selected from a normal population. The sample mean was 407 and the sample standard deviation 6. Using the .01 significance level:

- State the decision rule.
- Compute the value of the test statistic.
- What is your decision regarding the null hypothesis?

- The Rocky Mountain district sales manager of Rath Publishing Inc., a college textbook publishing company, claims that the sales representatives make an average of 40 sales calls per week on professors. Several reps say that this estimate is too low. To investigate, a random sample of 28 sales representatives reveals that the mean number of calls made last week was 42. The standard deviation of the sample is 2.1 calls. Using the .05 significance level, can we conclude that the mean number of calls per salesperson per week is more than 40?
- The management of White Industries is considering a new method of assembling its golf cart. The present method requires a mean time of 42.3 minutes to assemble a cart. The mean assembly time for a random sample of 24 carts, using the new method, was 40.6 minutes, and the standard deviation of the sample was 2.7 minutes. Using the .10 level of significance, can we conclude that the assembly time using the new method is faster?

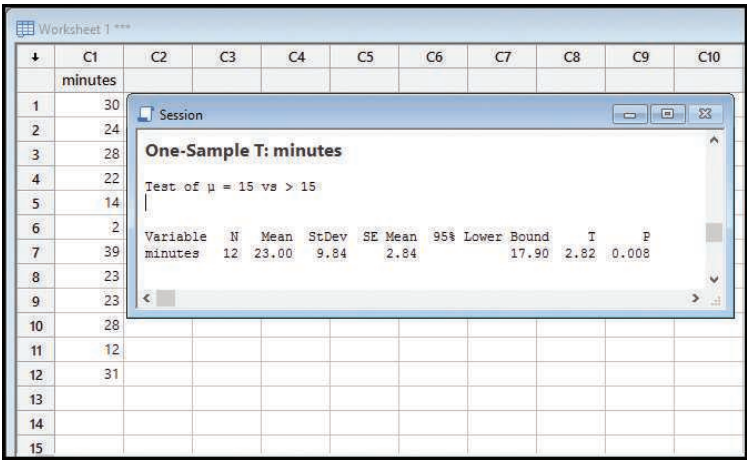
13. The mean income per person in the United States is \$50,000, and the distribution of incomes follows a normal distribution. A random sample of 10 residents of Wilmington, Delaware, had a mean of \$60,000 with a standard deviation of \$10,000. At the .05 level of significance, is that enough evidence to conclude that residents of Wilmington, Delaware, have more income than the national average?
14. **FILE** Most air travelers now use e-tickets. Electronic ticketing allows passengers to not worry about a paper ticket, and it costs the airline companies less to handle than paper ticketing. However, in recent times the airlines have received complaints from passengers regarding their e-tickets, particularly when connecting flights and a change of airlines were involved. To investigate the problem, an independent watchdog agency contacted a random sample of 20 airports and collected information on the number of complaints the airport had with e-tickets for the month of March. The information is reported below.

14	14	16	12	12	14	13	16	15	14
12	15	15	14	13	13	12	13	10	13

- At the .05 significance level, can the watchdog agency conclude the mean number of complaints per airport is less than 15 per month?
- a. What assumption is necessary before conducting a test of hypothesis?
 - b. Plot the number of complaints per airport in a frequency distribution or a dot plot. Is it reasonable to conclude that the population follows a normal distribution?
 - c. Conduct a test of hypothesis and interpret the results.

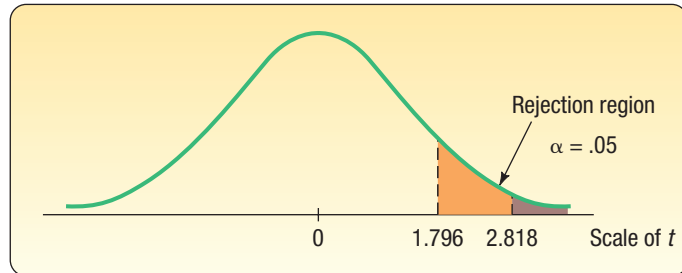
A Statistical Software Solution

The Minitab statistical software system, used in earlier chapters and the previous section, provides an efficient method for conducting a one-sample test of hypothesis for a population mean. The steps to generate the following output are shown in Appendix C.



An additional feature of most statistical software packages is to report the *p*-value, which gives additional information on the null hypothesis. The *p*-value is the probability of a *t* value as extreme or more extreme than the computed *t* value, given that the null hypothesis is true. Using the Minitab analysis from the previous cell phone parking lot example, the *p*-value of .008 is the likelihood of a *t* value of 2.82 or larger, given a population mean of 15. Thus, comparing the *p*-value to the significance level tells us whether the null hypothesis was close to being rejected, barely rejected, and so on.

To explain further, refer to the diagram below. The p -value of .008 is the brown shaded area and the significance level is the total amber and brown shaded area. Because the p -value of .008 is less than the significance level of .05, the null hypothesis is rejected. Had the p -value been larger—say, .06, .19, or .57—than the significance level, the null hypothesis would not be rejected.



In the preceding example, the alternate hypothesis was one-sided, and the upper (right) tail of the t distribution contained the rejection region. The p -value is the area to the right of 2.818 for a t distribution with 11 degrees of freedom.

What if we were conducting a two-sided test, so that the rejection region is in both the upper and the lower tails? That is, in the cell phone parking lot example, if H_1 were stated as $\mu \neq 15$, we would have reported the p -value as the area to the right of 2.818 plus the value to the left of -2.818 . Both of these values are .008, so the p -value is $.008 + .008 = .016$.

How can we estimate a p -value without a computer? To illustrate, recall that, in the example/solution regarding the length of time at the cell phone parking lot, we rejected the null hypothesis that $\mu \leq 15$ and accepted the alternate hypothesis that $\mu > 15$. The significance level was .05, so logically the p -value is less than .05. To estimate the p -value more accurately, go to Appendix B.5 and find the row with 11 degrees of freedom. The computed t value of 2.818 is between 2.718 and 3.106. (A portion of Appendix B.5 is reproduced as Table 10–3.) The one-tailed significance level corresponding to 2.718 is .01, and for 3.106 it is .005. Therefore, the p -value is between .005 and .01. The usual practice is to report that the p -value is less than the larger of the two significance levels. So we would report “the p -value is less than .01.”

TABLE 10–3 A Portion of Student’s t Distribution

Confidence Intervals						
	80%	90%	95%	98%	99%	99.9%
df	Level of Significance for One-Tailed Test, α					
	0.10	0.05	.0025	0.01	0.005	0.0005
	Level of Significance for Two-Tailed Test, α					
	0.20	0.10	0.05	0.02	0.01	0.001
∴	∴	∴	∴	∴	∴	∴
9	1.383	1.833	2.262	2.821	3.250	4.781
10	1.372	1.812	2.228	2.764	3.169	4.587
11	1.363	1.796	2.201	2.718	3.106	4.437
12	1.356	1.782	2.179	2.681	3.055	4.318
13	1.350	1.771	2.160	2.650	3.012	4.221
14	1.345	1.761	2.145	2.624	2.977	4.140
15	1.341	1.753	2.131	2.602	2.947	4.073

SELF-REVIEW 10-4



A machine is set to fill a small bottle with 9.0 grams of medicine. A sample of eight bottles revealed the following amounts (grams) in each bottle.

9.2 8.7 8.9 8.6 8.8 8.5 8.7 9.0

At the .01 significance level, can we conclude that the mean weight is less than 9.0 grams?

- State the null hypothesis and the alternate hypothesis.
- How many degrees of freedom are there?
- Give the decision rule.
- Compute the value of t . What is your decision regarding the null hypothesis?
- Estimate the p -value.

EXERCISES

15. Given the following hypotheses:

$$H_0: \mu \geq 20$$

$$H_1: \mu > 20$$

A random sample of five resulted in the following values: 18, 15, 12, 19, and 21. Assume a normal population. Using the .01 significance level, can we conclude the population mean is less than 20?

- State the decision rule.
- Compute the value of the test statistic.
- What is your decision regarding the null hypothesis?
- Estimate the p -value.

16. Given the following hypotheses:

$$H_0: \mu = 100$$

$$H_1: \mu \neq 100$$

A random sample of six resulted in the following values: 118, 105, 112, 119, 105, and 111. Assume a normal population. Using the .05 significance level, can we conclude the mean is different from 100?

- State the decision rule.
- Compute the value of the test statistic.
- What is your decision regarding the null hypothesis?
- Estimate the p -value.

17. **FILE** The amount of water consumed each day by a healthy adult follows a normal distribution with a mean of 1.4 liters. A health campaign promotes the consumption of at least 2.0 liters per day. A sample of 10 adults after the campaign shows the following consumption in liters:

1.5 1.6 1.5 1.4 1.9 1.4 1.3 1.9 1.8 1.7

At the .01 significance level, can we conclude that water consumption has increased? Calculate and interpret the p -value.

18. **FILE** The liquid chlorine added to swimming pools to combat algae has a relatively short shelf life before it loses its effectiveness. Records indicate that the mean shelf life of a 5-gallon jug of chlorine is 2,160 hours (90 days). As an experiment, Holdlonger was added to the chlorine to find whether it would increase the shelf life. A sample of nine jugs of chlorine had these shelf lives (in hours):

2,159 2,170 2,180 2,179 2,160 2,167 2,171 2,181 2,185

At the .025 level, has Holdlonger increased the shelf life of the chlorine? Estimate the p -value.

19. **FILE** A Washington, D.C., “think tank” announces the typical teenager sent 67 text messages per day in 2017. To update that estimate, you phone a sample of 12 teenagers and ask them how many text messages they sent the previous day. Their responses were:

51 175 47 49 44 54 145 203 21 59 42 100

At the .05 level, can you conclude that the mean number is greater than 67? Estimate the p -value and describe what it tells you.

20. **FILE** Hugger Polls contends that an agent conducts a mean of 53 in-depth home surveys every week. A streamlined survey form has been introduced, and Hugger wants to evaluate its effectiveness. The number of in-depth surveys conducted during a week by a random sample of 15 agents are:

53 57 50 55 58 54 60 52 59 62 60 60 51 59 56

At the .05 level of significance, can we conclude that the mean number of interviews conducted by the agents is more than 53 per week? Estimate the p -value.

LO10-7

Compute the probability of a Type II error.

TYPE II ERROR

Recall that the level of significance, identified by the symbol α , is the probability that the null hypothesis is rejected when it is true. This is called a Type I error. The most common levels of significance are .05 and .01 and are set by the researcher at the outset of the test.

In a hypothesis-testing situation there is also the possibility that a null hypothesis is not rejected when it is actually false. This is called a Type II error. The probability of a Type II error is identified by the Greek letter beta (β). In contrast to selecting a value for α in the hypothesis testing procedure, the value of β is calculated after the hypothesis testing procedure is finished. The following example illustrates the details of determining the value of β .

EXAMPLE

Western Wire Products purchases steel bars to make cotter pins. Past experience indicates that the mean tensile strength of all incoming shipments is 10,000 psi and that the standard deviation, σ , is 400 psi. To monitor the quality of the cotter pins, samples of 100 pins are randomly selected and tested for their strength. In our hypothesis testing procedure the hypotheses are:

$$H_0: \mu = 10,000$$

$$H_1: \mu \neq 10,000$$

To determine if a shipment of steel bars meets the quality standard, Western Wire Products set up a rule for the quality-control inspector to follow: “Take a sample of 100 steel bars. Test each of the bars for tensile strength. Using a .05 significance level, accept the shipment if the sample mean ($\bar{x}$) strength falls between 9,922 psi and 10,078 psi.” These values are the critical values for the hypothesis test. If the sample mean is more than 10,078 or less than 9,922, the hypothesis is rejected and we conclude that the shipment does not meet the quality standard.

Refer to Chart 10–8, Graph A. Given that the population mean is 10,000 psi, designated μ_0 , with a standard deviation of 400, the distribution shows the regions where the hypothesis is rejected and where it is not rejected, that is, whether the shipment meets the quality standard for tensile strength.

Suppose that the result of testing 100 bars results in a sample mean of 9,900 psi. Clearly, based on Graph A, the shipment does not meet the quality standard and is rejected with a .05 probability of a Type I error, a small chance of rejecting the shipment in error. To calculate the Type II error, we assume that the sample mean of 9,900 psi is the true population mean. Compare Graph A and Graph B. Graph A represents the company's distribution of tensile strength centered on 10,000 psi. Graph B, based on the sample data, suggests that the distribution is centered on 9,900 psi.

Now, let's use Graph B to determine the probability of a Type II error, β . From the quality standards we know that 9,922 psi is used to reject the null hypothesis. Any sample mean greater than 9,922 and less than 10,078 is accepted. If the distribution is really centered on 9,900 psi, it is possible to find sample means more than 9,922, and we would fail to reject the null hypothesis, $\mu = 10,000$. This is the area in Graph B labeled "Probability of β " where the null hypothesis would not be rejected. The graph indicates that the probability of a Type II error is .2912. We can also calculate the power of the test as $(1 - \beta)$. The power is the probability of not making a Type II error, rejecting the null hypothesis correctly. Here, the power of the test is .7088.

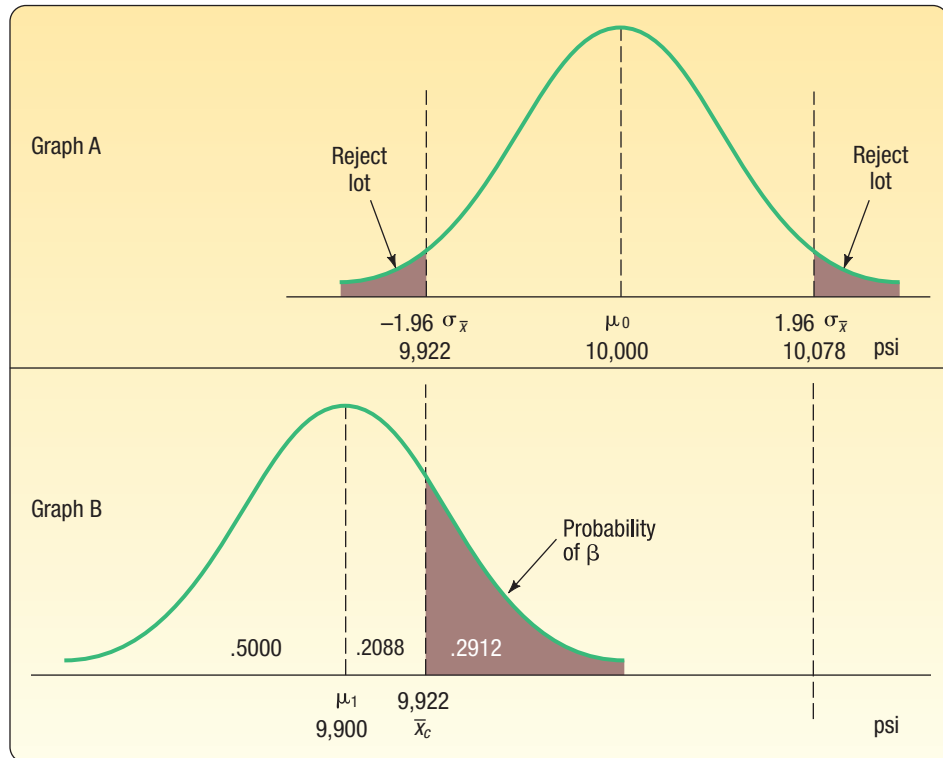


CHART 10-8 Charts Showing Type I and Type II Errors

SOLUTION

The probability of committing a Type II error, is represented by the shaded area in Chart 10-8, Graph B. It is computed by determining the area under the normal curve that lies above 9,922 pounds. The calculation of the areas under the normal curve was discussed in Chapter 7. Reviewing briefly, first determine the probability of the sample mean falling between 9,900 and 9,922. Then this probability is subtracted from .5000 (which represents all the area to the right of the mean of 9,900) to arrive at the probability of making a Type II error.

The number of standard errors (z value) between the mean of the incoming lot (9,900), designated by μ_1 , and $\bar{x}_c$, representing the critical value for 9,922, is computed by:

$$\text{TYPE II ERROR} \quad z = \frac{\bar{x}_c - \mu_1}{\sigma/\sqrt{n}} \quad (10-3)$$

With $n = 100$ and $\sigma = 400$, the value of z is 0.55:

$$z = \frac{\bar{x}_c - \mu_1}{\sigma/\sqrt{n}} = \frac{9,922 - 9,900}{400/\sqrt{100}} = \frac{22}{40} = 0.55$$

The area under the curve between 9,900 and 9,922 (a z value of 0.55) is .2088. The area under the curve beyond 9,922 pounds is .5000 – .2088, or .2912; this is the probability of making a Type II error—that is, accepting an incoming lot of steel bars, based on the company's standard of 10,000 psi, when the sample suggests that the population mean is 9,900 psi and the shipment should be rejected.

Let's use another example, illustrated in Chart 10–9. Suppose the testing of a sample of 100 bars results in a mean of 10,120 psi. Based on the company's standards, represented in Graph A, the shipment should be rejected. The sample mean of 10,120 is more than the critical value of 10,078. Based on the sample mean, Graph C shows the distribution of sample means based on the distribution centered on the sample mean, 10,120.

If the distribution is centered on the sample mean of 10,120, there is a probability that a sample mean could be less than the upper critical value of 10,078. In Graph C, this is labeled, β . To find the probability, we calculate the z value of 10,078 in Graph C:

$$z = \frac{\bar{x}_c - \mu_1}{\sigma/\sqrt{n}} = \frac{10,078 - 10,120}{400/\sqrt{100}} = -1.05$$

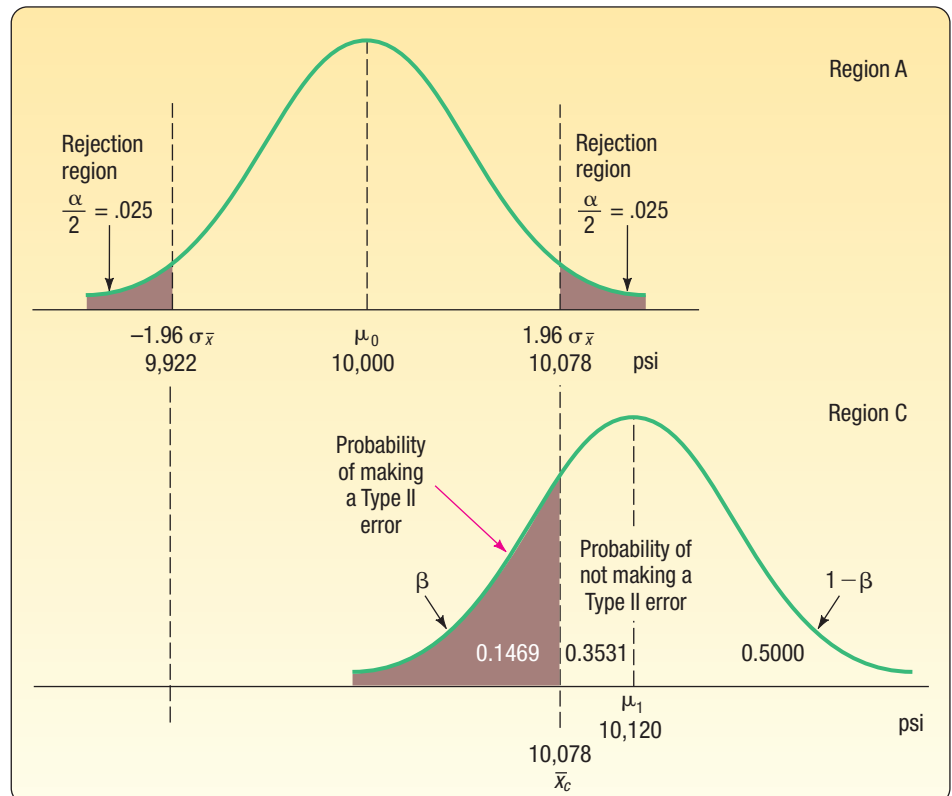


CHART 10–9 Type I and Type II Errors (Another Example)

The probability that z is less than -1.05 is $.1469$, found by $.5000 - .3531$. Therefore, β , or the probability of a Type II error, is $.1469$. The power of the test is $1.000 - .1469$ or $.8531$. Caution: If the difference between μ_0 and μ_1 is relatively small, the probability of a Type II error may occur in both tails. This eventuality is not considered here.

Using the methods illustrated by Charts 10–8 Region B and 10–9 Region C, the probability of a Type II error can be determined for any sample mean used as an estimate of μ_1 .

To review, in hypothesis testing, we select the probability of making a Type I error, α . Naturally, we pick small probabilities, typically less than 0.10 . The probability of a Type II error depends on the sample result. The larger the difference between the hypothesized mean and the sample mean, the smaller the probability of a Type II error, failing to reject the null hypothesis when it should have been rejected.

For this example/solution, Table 10–4 shows the probabilities of a Type II error and the power of the test for selected values of μ_1 . The right column gives the probability of not making a Type II error, which is also known as the power of a test.

TABLE 10–4 Probabilities of a Type II Error for $\mu_0 = 10,000$ Pounds and Selected Alternative Means, .05 Level of Significance

Selected Mean (μ_1)	Probability of Type II Error (β)	Power of a Test ($1 - \beta$)
9,820	.0054	.9946
9,880	.1469	.8531
9,900	.2912	.7088
9,940	.6736	.3264
10,000	— *	—
10,060	.6736	.3264
10,100	.2912	.7088
10,120	.1469	.8531
10,180	.0054	.9946

*It is not possible to make a Type II error when $\mu_1 = \mu_0$.

SELF-REVIEW 10–5



Refer to the previous example. Suppose the true mean of an incoming lot of steel bars is 10,180 psi. What is the probability that the quality control inspector will accept the bars as having a mean of 10,000 psi? (It sounds implausible that steel bars will be rejected if the tensile strength is higher than specified. However, it may be that the cotter pin has a dual function in an outboard motor. It may be designed not to shear off if the motor hits a small object, but to shear off if it hits a rock. Therefore, the steel should not be *too* strong.)

The light area in Chart 10–9, Region C, represents the probability of falsely accepting the hypothesis that the mean tensile strength of the incoming steel is 10,000 psi. What is the probability of committing a Type II error?

EXERCISES

- 21. Refer to Table 10–4 and the example just completed. With $n = 100$, $\sigma = 400$, $\bar{x}_c = 9,922$ and $\mu_1 = 9,880$, verify that the probability of a Type II error is $.1469$.
- 22. Refer to Table 10–4 and the example just completed. With $n = 100$, $\sigma = 400$, $\bar{x}_c = 10,078$, and $\mu_1 = 10,100$, verify that the probability of a Type II error is $.2912$.
- 23. The management of KSmall Industries is considering a new method of assembling a computer. The current assembling method requires a mean time of 60 minutes

with a standard deviation of 2.7 minutes. Using the new method, the mean assembly time for a random sample of 24 computers was 58 minutes. Using the .10 level of significance, can we conclude that the assembly time using the new method is faster? What is the probability of a Type II error?

24. A recent national survey found that parents read an average (mean) of 10 books per month to their children under five years old. The population standard deviation is 5. The distribution of books read per month follows the normal distribution. A random sample of 25 households revealed that the mean number of books read last month was 12. At the .01 significance level, can we conclude that parents read more than the average number of books to their children? What is the probability of a Type II error?

CHAPTER SUMMARY

- I. The objective of hypothesis testing is to verify the validity of a statement about a population parameter.
- II. The steps to conduct a test of hypothesis are:
 - A. State the null hypothesis (H_0) and the alternate hypothesis (H_1).
 - B. Select the level of significance.
 1. The level of significance is the likelihood or probability of rejecting a true null hypothesis.
 2. The most frequently used probabilities used as significance levels are .01, .05, and .10. As a probability, any value between 0 and 1.00 is possible, but we prefer small probabilities of making a Type I error.
 - C. Select the test statistic.
 1. A test statistic is a value calculated from sample information used to determine whether to reject the null hypothesis.
 2. Two test statistics were considered in this chapter.
 - a. The standard normal distribution (the z distribution) is used when the population follows the normal distribution and the population standard deviation is known.
 - b. The t distribution is used when the population follows the normal distribution and the population standard deviation is unknown.
 - D. State the decision rule.
 1. The decision rule indicates the condition or conditions when the null hypothesis is rejected.
 2. In a two-tailed test, the rejection region is evenly split between the upper and lower tails.
 3. In a one-tailed test, all of the rejection region is in either the upper or the lower tail.
 - E. Select a sample, compute the value of the test statistic, and make a decision regarding the null hypothesis.
 - F. Interpret the results of your decision.
- III. A p -value is the probability that the value of the test statistic is as extreme as the value computed, when the null hypothesis is true.
- IV. When testing a hypothesis about a population mean:
 - A. If the population standard deviation, σ , is known, the test statistic is the standard normal distribution and is determined from:

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \quad (10-1)$$

- B. If the population standard deviation is not known, s is substituted for σ . The test statistic is the t distribution, and its value is determined from:

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} \quad (10-2)$$

- The major characteristics of the t distribution are:
- 1. It is a continuous distribution.
 - 2. It is mound-shaped and symmetrical.
 - 3. It is flatter, or more spread out, than the standard normal distribution.
 - 4. There is a family of t distributions, depending on the number of degrees of freedom.
- V. There are two types of errors that can occur in a test of hypothesis.
- A. A Type I error occurs when a true null hypothesis is rejected.
 - 1. The probability of making a Type I error is equal to the level of significance.
 - 2. This probability is designated by the Greek letter α .
 - B. A Type II error occurs when a false null hypothesis is not rejected.
 - 1. The probability of making a Type II error is designated by the Greek letter β .
 - 2. The likelihood of a Type II error must be calculated comparing the hypothesized distribution to an alternate distribution based on sample results.

PRONUNCIATION KEY

SYMBOL	MEANING	PRONUNCIATION
H_0	Null hypothesis	<i>H sub zero</i>
H_1	Alternate hypothesis	<i>H sub one</i>
$\alpha/2$	Two-tailed significance level	<i>Alpha divided by 2</i>
$\bar{x}_c$	Limit of the sample mean	<i>x bar sub c</i>
μ_0	Assumed population mean	<i>mu sub zero</i>

CHAPTER EXERCISES

25. According to the local union president, the mean gross income of plumbers in the Salt Lake City area follows the normal probability distribution with a mean of \$45,000 and a standard deviation of \$3,000. A recent investigative reporter for KYAK TV found, for a sample of 120 plumbers, the mean gross income was \$45,500. At the .10 significance level, is it reasonable to conclude that the mean income is not equal to \$45,000? Determine the p -value.
26. **FILE** Rutter Nursery Company packages its pine bark mulch in 50-pound bags. From a long history, the production department reports that the distribution of the bag weights follows the normal distribution and the standard deviation of the packaging process is 3 pounds per bag. At the end of each day, Jeff Rutter, the production manager, weighs 10 bags and computes the mean weight of the sample. Below are the weights of 10 bags from today's production.
- 45.6 47.7 47.6 46.3 46.2 47.4 49.2 55.8 47.5 48.5
- a. Can Mr. Rutter conclude that the mean weight of the bags is less than 50 pounds? Use the .01 significance level.
 - b. In a brief report, tell why Mr. Rutter can use the z distribution as the test statistic.
 - c. Compute the p -value.
27. A new weight-watching company, Weight Reducers International, advertises that those who join will lose an average of 10 pounds after the first two weeks. The standard deviation is 2.8 pounds. A random sample of 50 people who joined the weight reduction program revealed a mean loss of 9 pounds. At the .05 level of significance, can we conclude that those joining Weight Reducers will lose less than 10 pounds? Determine the p -value.
28. Dole Pineapple Inc. is concerned that the 16-ounce can of sliced pineapple is being overfilled. Assume the standard deviation of the process is .03 ounce. The quality-control department took a random sample of 50 cans and found that the arithmetic mean weight was 16.05 ounces. At the 5% level of significance, can we conclude that the mean weight is greater than 16 ounces? Determine the p -value.

29. According to a recent survey, Americans get a mean of 7 hours of sleep per night. A random sample of 50 students at West Virginia University revealed the mean length of time slept last night was 6 hours and 48 minutes (6.8 hours). The standard deviation of the sample was 0.9 hour. At the 5% level of significance, is it reasonable to conclude that students at West Virginia sleep less than the typical American? Compute the p -value.
30. A statewide real estate sales agency, Farm Associates, specializes in selling farm property in the state of Nebraska. Its records indicate that the mean selling time of farm property is 90 days. Because of recent drought conditions, the agency believes that the mean selling time is now greater than 90 days. A statewide survey of 100 recently sold farms revealed a mean selling time of 94 days, with a standard deviation of 22 days. At the .10 significance level, has there been an increase in selling time?
31. According to the Census Bureau, 3.13 people reside in the typical American household. A sample of 25 households in Arizona retirement communities showed the mean number of residents per household was 2.86 residents. The standard deviation of this sample was 1.20 residents. At the .05 significance level, is it reasonable to conclude the mean number of residents in the retirement community household is less than 3.13 persons?
32. A recent article in *Vitality* magazine reported that the mean amount of leisure time per week for American men is 40.0 hours. You believe this figure is too large and decide to conduct your own test. In a random sample of 60 men, you find that the mean is 37.8 hours of leisure per week and that the standard deviation of the sample is 12.2 hours. Can you conclude that the information in the article is untrue? Use the .05 significance level. Determine the p -value and explain its meaning.
33. **FILE** A recent survey by *nerdwallet.com* indicated Americans paid a mean of \$6,658 interest on credit card debt in 2017. A sample of 12 households with children revealed the follow amounts. At the .05 significance level is it reasonable to conclude that these households paid more interest?

7077 5744 6753 7381 7625 6636 7164 7348 8060 5848 9275 7052

34. **FILE** A recent article in *The Wall Street Journal* reported that the home equity loan rate is now less than 4%. A sample of eight small banks in the Midwest revealed the following home equity loan rates (in percent):

3.6 4.1 5.3 3.6 4.9 4.6 5.0 4.4

At the .01 significance level, can we conclude that the home equity loan rate for small banks is less than 4%? Estimate the p -value.

35. **FILE** A recent study revealed the typical American coffee drinker consumes an average of 3.1 cups per day. A sample of 12 senior citizens revealed they consumed the following amounts of coffee, reported in cups, yesterday.

3.1 3.3 3.5 2.6 2.6 4.3 4.4 3.8 3.1 4.1 3.1 3.2

At the .05 significance level, do these sample data suggest there is a difference between the national average and the sample mean from senior citizens?

36. **FILE** The postanesthesia care area (recovery room) at St. Luke's Hospital in Maumee, Ohio, was recently enlarged. The hope was that the change would increase the mean number of patients served per day to more than 25. A random sample of 15 days revealed the following numbers of patients.

25 27 25 26 25 28 28 27 24 26 25 29 25 27 24

At the .01 significance level, can we conclude that the mean number of patients per day is more than 25? Estimate the p -value and interpret it.

37. **FILE** www.golfsmith.com receives an average of 6.5 returns per day from online shoppers. For a sample of 12 days, it received the following numbers of returns.

0	4	3	4	9	4	5	9	1	6	7	10
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At the .01 significance level, can we conclude the mean number of returns is less than 6.5?

38. **FILE** During recent seasons, Major League Baseball has been criticized for the length of the games. A report indicated that the average game lasts 3 hours and 30 minutes. A sample of 17 games revealed the following times to completion. (Note that the minutes have been changed to fractions of hours, so that a game that lasted 2 hours and 24 minutes is reported at 2.40 hours.)

2.98	2.40	2.70	2.25	3.23	3.17	2.93	3.18	2.80
2.38	3.75	3.20	3.27	2.52	2.58	4.45	2.45	

Can we conclude that the mean time for a game is less than 3.50 hours? Use the .05 significance level.

39. **FILE** Watch Corporation of Switzerland claims that its watches on average will neither gain nor lose time during a week. A sample of 18 watches provided the following gains (+) or losses (−) in seconds per week.

−0.38	−0.20	−0.38	−0.32	+0.32	−0.23	+0.30	+0.25	−0.10
−0.37	−0.61	−0.48	−0.47	−0.64	−0.04	−0.20	−0.68	+0.05

Is it reasonable to conclude that the mean gain or loss in time for the watches is 0? Use the .05 significance level. Estimate the p -value.

40. **FILE** Listed below is the annual rate of return (reported in percent) for a sample of 12 taxable mutual funds.

4.63	4.15	4.76	4.70	4.65	4.52	4.70	5.06	4.42	4.51	4.24	4.52
------	------	------	------	------	------	------	------	------	------	------	------

Using the .05 significance level, is it reasonable to conclude that the mean rate of return is more than 4.50%?

41. **FILE** Many grocery stores and large retailers such as Kroger and Walmart have installed self-checkout systems so shoppers can scan their own items and cash out themselves. How do customers like this service and how often do they use it? Listed below is the number of customers using the service for a sample of 15 days at a Walmart location.

120	108	120	114	118	91	118	92	104	104
112	97	118	108	117					

Is it reasonable to conclude that the mean number of customers using the self-checkout system is more than 100 per day? Use the .05 significance level.

42. **FILE** For a recent year, the mean fare to fly from Charlotte, North Carolina, to Chicago, Illinois, on a discount ticket was \$267. A random sample of 13 round-trip discount fares on this route last month shows:

\$321	\$286	\$290	\$330	\$310	\$250	\$270	\$280	\$299	\$265	\$291	\$275	\$281
-------	-------	-------	-------	-------	-------	-------	-------	-------	-------	-------	-------	-------

At the .01 significance level, can we conclude that the mean fare has increased? What is the p -value?

43. The publisher of *Celebrity Living* claims that the mean sales for personality magazines that feature people such as Megan Fox or Jennifer Lawrence are 1.5 million copies per week. A sample of 10 comparable titles shows a mean weekly sales last week of 1.3 million copies with a standard deviation of 0.9 million copies. Do these data contradict the publisher's claim? Use the 0.01 significance level.

44. A United Nations report shows the mean family income for Mexican migrants to the United States is \$27,000 per year. A FLOC (Farm Labor Organizing Committee) evaluation of 25 Mexican family units reveals a mean to be \$30,000 with a sample standard deviation of \$10,000. Does this information disagree with the United Nations report? Apply the 0.01 significance level.
45. **FILE** The number of “destination weddings” has skyrocketed in recent years. For example, many couples are opting to have their weddings in the Caribbean. A Caribbean vacation resort recently advertised in *Bride Magazine* that the cost of a Caribbean wedding was less than \$30,000. Listed below is a total cost in \$000 for a sample of 8 Caribbean weddings.

29.7	29.4	31.7	29.0	29.1	30.5	29.1	29.8
------	------	------	------	------	------	------	------

- At the .05 significance level, is it reasonable to conclude the mean wedding cost is less than \$30,000 as advertised?
46. The American Water Works Association reports that the per capita water use in a single-family home is 69 gallons per day. Legacy Ranch is a relatively new housing development. The builders installed more efficient water fixtures, such as low-flush toilets, and subsequently conducted a survey of the residences. Thirty-six owners responded, and the sample mean water use per day was 64 gallons with a standard deviation of 8.8 gallons per day. At the .10 level of significance, is that enough evidence to conclude that residents of Legacy Ranch use less water on average?
47. A cola-dispensing machine is set to dispense 9.00 ounces of cola per cup, with a standard deviation of 1.00 ounce. The manufacturer of the machine would like to set the control limit in such a way that, for samples of 36, 5% of the sample means will be greater than the upper control limit, and 5% of the sample means will be less than the lower control limit.
- At what value should the control limit be set?
 - If the population mean shifts to 8.6, what is the probability of detecting the change?
 - If the population mean shifts to 9.6, what is the probability of detecting the change?
48. The owners of the Westfield Mall wished to study customer shopping habits. From earlier studies, the owners were under the impression that a typical shopper spends 0.75 hour at the mall, with a standard deviation of 0.10 hour. Recently the mall owners added some specialty restaurants designed to keep shoppers in the mall longer. The consulting firm, Brunner and Swanson Marketing Enterprises, was hired to evaluate the effects of the restaurants. A sample of 45 shoppers by Brunner and Swanson revealed that the mean time spent in the mall had increased to 0.80 hour.
- Develop a hypothesis test to determine if the mean time spent in the mall changed. Use the .10 significance level.
 - Suppose the mean shopping time actually increased from 0.75 hour to 0.79 hours. What is the probability of making a Type II error?
 - When Brunner and Swanson reported the information in part (b) to the mall owners, the owners believed that the probability of making a Type II error was too high. How could this probability be reduced?
49. The following null and alternate hypotheses are given.

$$H_0: \mu \leq 50$$

$$H_1: \mu > 50$$

- Suppose the population standard deviation is 10. The probability of a Type I error is set at .01 and the probability of a Type II error at .30. Assume that the population mean shifts from 50 to 55. How large a sample is necessary to meet these requirements?
50. An insurance company, based on past experience, estimates the mean damage for a natural disaster in its area is \$5,000. After introducing several plans to prevent loss, it randomly samples 200 policyholders and finds the mean amount per claim was \$4,800 with a standard deviation of \$1,300. Does it appear the prevention plans were effective in reducing the mean amount of a claim? Use the .05 significance level.
51. A national grocer’s magazine reports the typical shopper spends 8 minutes in line waiting to check out. A sample of 24 shoppers at the local Farmer Jack’s showed a mean of 7.5 minutes with a standard deviation of 3.2 minutes. Is the waiting time at the local Farmer Jack’s less than that reported in the national magazine? Use the .05 significance level.

DATA ANALYTICS

52. **FILE** The North Valley Real Estate data reports information on the homes sold last year.
- Adam Marty recently joined North Valley Real Estate and was assigned twenty homes to market and show. When he was hired, North Valley assured him that the twenty homes would be fairly assigned to him. When he reviewed the selling prices of his assigned homes, he thought that the prices were much below the average of \$357,000. Adam was able to find the data of how the other agents in the firm were assigned to the homes. Use statistical inference to analyze the “fairness” that homes were assigned to the agents.
53. **FILE** Refer to the Baseball 2016 data, which report information on the 30 Major League Baseball teams for the 2016 season.
- Conduct a test of hypothesis to determine whether the mean salary of the teams was different from \$100.0 million. Use the .05 significance level.
 - Using a 5% significance level, conduct a test of hypothesis to determine whether the mean attendance was more than 2,000,000 per team.
54. **FILE** Refer to the Lincolnville School District bus data.
- Select the variable for the number of miles traveled last month. Conduct a hypothesis test to determine whether the mean miles traveled last month equals 10,000. Use the .01 significance level. Find the p -value and explain what it means.
 - A study of school bus fleets reports that the average per bus maintenance cost is \$4,000 per year. Using the maintenance cost variable, conduct a hypothesis test to determine whether the mean maintenance cost for Lincolnville’s bus fleet is more than \$4,000 at the .05 significance level. Determine the p -value and report the results.